

CERTIFICATE OF AUTHORSHIP

Instructions

- Download and fill this PDF form completely.
- Each course requirement submission, unless otherwise specified by the Course Instructor, whether in electronic or paper form, must be accompanied by a corresponding properly accomplished Certificate of Authorship.

Description of Submission

Title of Submission: Group Exercise Lab 4

Type of Submission: ☐ Program ☐ Project ☐ Report ☐ Paper
☒ Other (specify) PDF file, CPP file, S files

Date of Submission: February 12, 2026

Certification

We hereby certify that the submission described in this document abides by the principles stipulated in the DISCS Academic Integrity Policy document. We further certify that we are the authors of this submission and that any assistance we received in its preparation is fully acknowledged and disclosed in the documentation. We have also cited all sources from which we obtained data, ideas, or words that are directly copied or paraphrased in this document. Sources are properly credited according to accepted standards for professional publication.

https://www.cs.sfu.ca/~ashriram/Courses/CS295/assets/books/CSAPP_2016.pdf
<https://stackoverflow.com/questions/9083743/is-bit-shifting-o1-or-on#:~:text=You%20may%20think%20shifting%208,17>
<https://medium.com/@akumar54/how-compilers-optimize-multiplication-a-deep-dive-with-assembly-and-debugging-271a3aaa448d>
[https://www.phys.ufl.edu/~coldwell/MultiplePrecision/fpvsintmult.htm#:~:text=Central%20Processing%20Unit%20\(CPU\)%3A,bit%20integer%20multiply%3A%2020%20https://cs.brown.edu/courses/cs033/docs/guides/x64_cheatsheet.pdf](https://www.phys.ufl.edu/~coldwell/MultiplePrecision/fpvsintmult.htm#:~:text=Central%20Processing%20Unit%20(CPU)%3A,bit%20integer%20multiply%3A%2020%20https://cs.brown.edu/courses/cs033/docs/guides/x64_cheatsheet.pdf)
https://www.cs.cmu.edu/~cburch/211-sp96/gnu_devel/gpp.html

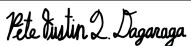
Declaration of Use of Generative AI

Tool:

Purpose:

We have reviewed and revised the content as we see fit. We take full responsibility for the content and ownership of the submitted / published work.

Group Information

Full Name	Signature	Course Code & Section
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